Assignment – 2 (GLM)

① Poisson distribution

$$E(k;\lambda) = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} \cdot k = \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1} e^{-\lambda}}{(k-1)!}$$

$$= \lambda$$

Let $p(y|\lambda) = \frac{\lambda^y e^{-\lambda}}{y!}$ (poisson distr).

General form of exponential family

$$= b(y) \exp(\eta T(y) - a(\eta))$$

$$p = \exp(y \log\lambda - \lambda) \cdot \frac{1}{y!}.$$

→ comparing, $\boxed{\frac{1}{y!} = b(y)}$

$\boxed{\eta = \log\lambda}$ $\boxed{a(\eta) = \lambda}$

$\therefore \boxed{a(\eta) = e^{\eta}}$ $\therefore \boxed{a'(\eta) = e^{\eta}}$

$\therefore$ Canonical response fn $\mu = g(\eta) = E(y|\eta)$

$= a'(\eta) = e^{\eta}$

in GLM, $\boxed{\eta = \theta^T x}$

$\boxed{\therefore \text{Canonical response fn} = e^{\eta}}$

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② we can write, ex:-

$$T(y) = \begin{vmatrix} 1 \\ 0 \\ \vdots \\ \vdots \end{vmatrix} \begin{matrix} \to \text{is } 1 \ (1 \text{ else } 0) \\ \to \text{is } 2 \ (1 \text{ else } 0) \\ \vdots \\ \vdots \end{matrix}$$

1 if $x = i$
0 otherwise

$[\,0] \rightarrow$ is k (1 else 0)

$$P(y \mid \{p_1, p_2 \cdots p_k\}) = \prod_{i=1}^{k} p_i^{[x=i]}$$

Let's write $[x=i]$ as $T_i(y)$,

Then $P(y \mid \{\ldots\}) = e^{\left(\sum_i T_i(y) \log p_i\right)}$

write p_i as $p = \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_k \end{bmatrix}$ But $\sum_i p_i =$

Then $p = (\log p)^T T(y) - \log \sum p_i$

Then, comparing, $\boxed{\log p = \eta}$

$a(\eta) = \log(\sum p_i) = \log(\sum e^{\eta})$.

Then, Taking gradient,

$$\boxed{\nabla a(\eta)^{(i)} = \frac{e^{\eta^{(i)}}}{\sum e^{\eta^{(i)}}}}$$

= expectation of $y \mid \eta$

$$\therefore \boxed{g(\eta)^{(i)} = E[y \mid \eta] = \frac{e^{\eta^{(i)}}}{\sum e^{\eta^{(i)}}}}$$

softmax

↳ canonical response fn

here, there is one extra parameter, one way is to make $\theta_k = 0$, then output only first $k-1$ as $h_\theta(x)$ and find the other by subtracting.

} CS229 lecture notes.

$\eta = \theta^T x \qquad \theta^T = k \times n$

features in x

for glm.

no. of features : x.

$$X = \underbrace{n \times 1}$$